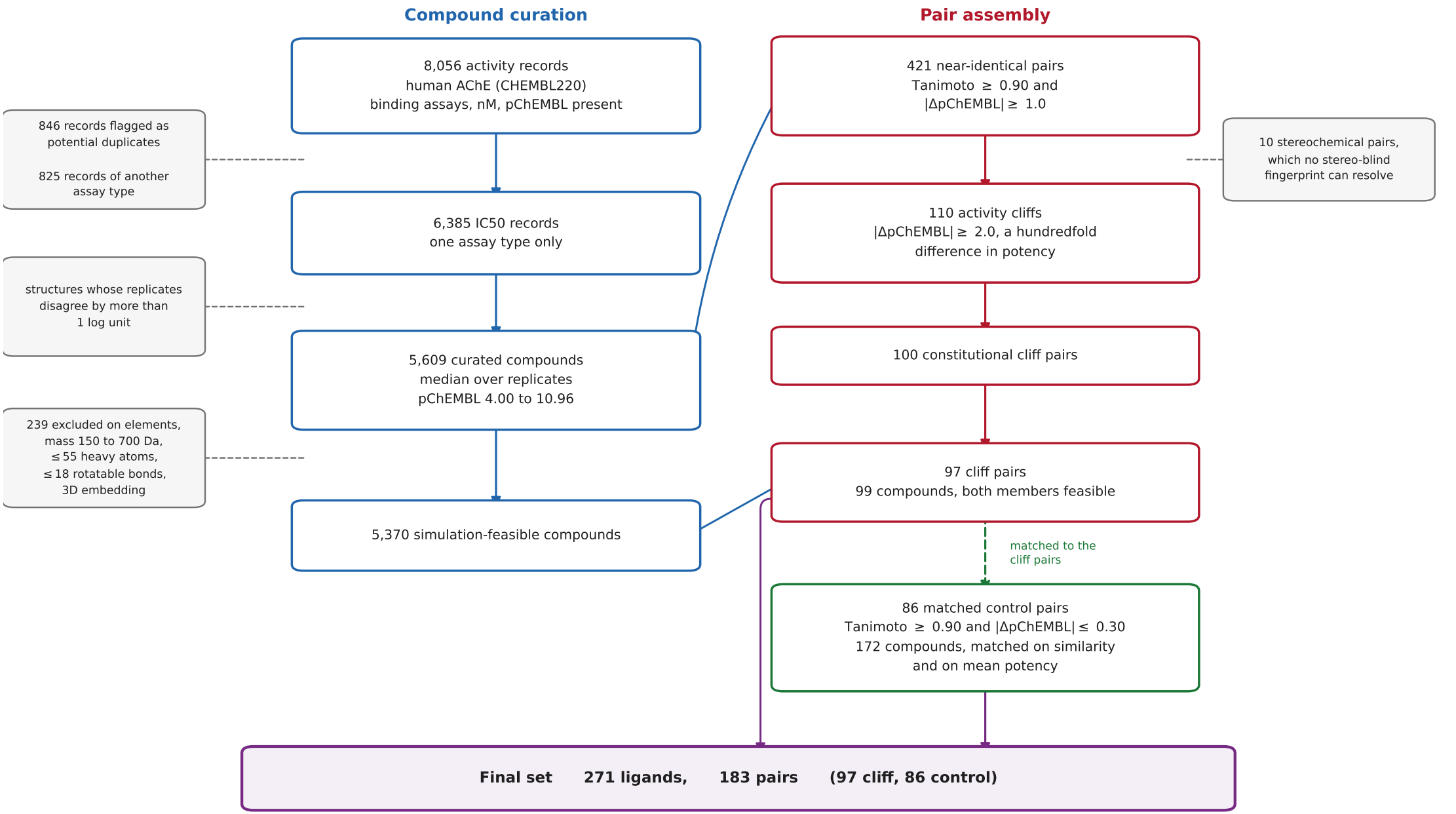
Compound curation
Pair assembly
8,056 activity records
human AChE (CHEMBL220)
binding assays, nM, pChEMBL present
846 records flagged as
potential duplicates
825 records of another
assay type
6,385 IC50 records
one assay type only
structures whose replicates
disagree by more than
1 log unit
5,609 curated compounds
median over replicates
pChEMBL 4.00 to 10.96
239 excluded on elements,
mass 150 to 700 Da,
≤ 55 heavy atoms,
≤ 18 rotatable bonds,
3D embedding
5,370 simulation-feasible compounds
421 near-identical pairs
Tanimoto ≥ 0.90 and
|ΔpChEMBL| ≥ 1.0
10 stereochemical pairs,
which no stereo-blind
fingerprint can resolve
110 activity cliffs
|ΔpChEMBL| ≥ 2.0, a hundredfold
difference in potency
100 constitutional cliff pairs
97 cliff pairs
99 compounds, both members feasible
matched to the
cliff pairs
86 matched control pairs
Tanimoto ≥ 0.90 and |ΔpChEMBL| ≤ 0.30
172 compounds, matched on similarity
and on mean potency
Final set 271 ligands, 183 pairs (97 cliff, 86 control)